# MRP-LLM: Multitask Reflective Large Language Models for Privacy-Preserving Next POI Recommendation


## Abstract

Large language models (LLMs) have shown promising potential for next Point-of-Interest (POI) recommendation. However, existing methods only perform direct zero-shot prompting, leading to ineffective extraction of user preferences, insufficient injection of collaborative signals, and a lack of user privacy protection. As such, we propose a novel **M**ultitask **R**eflective **L**arge **L**anguage **M**odel for **P**rivacy-preserving Next POI Recommendation (MRP-LLM), aiming to exploit LLMs for better next POI recommendation while preserving user privacy. Specifically, the *Multitask Reflective Preference Extraction Module* first utilizes LLMs to distill each user's fine-grained (i.e., categorical, temporal, and spatial) preferences into a knowledge base (KB). The *Neighbor Preference Retrieval Module* retrieves and summarizes the preferences of similar users from the KB to obtain collaborative signals. Subsequently, aggregating the user's preferences with those of similar users, the *Multitask Next POI Recommendation Module* generates the next POI recommendations via multitask prompting. Meanwhile, during data collection, a *Privacy Transmission Module* is specifically devised to preserve sensitive POI data. Extensive experiments on three real-world datasets demonstrate the efficacy of MRP-LLM in providing more accurate next POI recommendations with user privacy preserved.



## Keywords

Next POI Recommendation, Large Language Models, Privacy Preservation, Responsible AI


## 1 Introduction

Next Point-of-Interest (POI) recommendation has been widely deployed in location social networks such as Yelp and Foursquare to help users discover relevant places and reduce information overload [44]. It is becoming a form of everyday decision support that shapes where people go, how they allocate time, and how they experience their cities. The resulting challenge is how to quickly identify a small set of options that align with one's current intent, while still reflecting personal preferences under situational constraints of time and distance. What is more, as the service increasingly relies on sensitive user data, including users' check-in histories, behavioral patterns, demographic and social attributes, ensuring that recommendation algorithms operate in an effective, responsible, and safe manner becomes critical to promote user experience and trust.

Conventional next POI recommender systems (RSs) usually train a domain-specific model based on users' check-in sequences to identify users' preferences on locations[1]. In pursuit of higher accuracy, next POI RSs have evolved from Markov-chain-based methods [42] and RNN-based methods [29] to transformer-based methods [35] and graph-based methods [27]. Despite the success, they still face challenges, including data sparsity and cold-start users. Fortunately, recent advancements in large language models (LLMs) such as ChatGPT have presented new opportunities to overcome the limitations

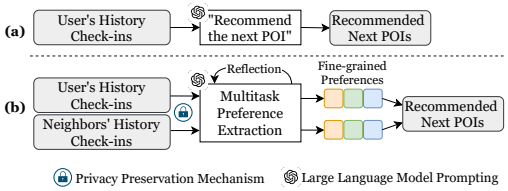


**Figure 1: Our method in (b), which makes recommendations based on diverse, fine-grained, and protected user preferences, compared to existing direct zero-shot methods in (a).**

and further enhance next POI RSs. First, LLMs have demonstrated the ability to transfer knowledge, such as the category knowledge of POIs, from the training corpus of other domains [17], which could facilitate users' POI preferences learning even with sparse user data. Second, in-context learning (ICL) shows strong potential for next-item recommendation, enabling robust learning even for cold start users [61]. Third, due to their natural language reasoning capabilities, LLMs can improve the interpretability of the recommendations, thereby enhancing users' trust in the system.

Despite their remarkable achievements across various recommendation domains [53, 54], the application of LLMs to next POI recommendation has only recently begun to receive attention. While existing work employs zero-shot prompting to extract users' spatial-temporal preferences for POIs [12, 13, 60], it exhibits several inherent shortcomings. **(1)** *Ineffective extraction of user preferences.* Successful POI RSs usually focus on accurately capturing users' POI-related preferences [67]. Instead, LLM-based next POI RSs lack a clear reasoning for the impact of fine-grained user preferences on their decisions. Thus, they have shown weaknesses in understanding the spatial-temporal information of user check-ins and mining user transition preferences, which can undermine the accuracy [13]. **(2)** *Insufficient injection of collaborative signals.* Conventional next POI RSs use collaborative signals by analyzing behaviors of similar users to obtain global behavioral patterns and enhance accuracy and generalizability. In contrast, existing LLM-based methods rely exclusively on the user's own check-in history, resulting in a limited view that lacks insights from broader datasets. **(3)** *Lacking of privacy protection.* The existing LLM-based methods directly expose user check-in history to LLMs, which may contain sensitive information such as users' whereabouts, home addresses, and behavioral habits [1]. A lack of protection for sensitive user data and the absence of quantitative privacy guarantees against potential information leakage pose a risk to users' trust. From users' perspective, LLM-based next-POI RSs should support three user-facing goals: (i) *Perceived relevance* under changing context and preference drift, (ii) *Transparency and contestability* through understandable preference-grounded explanations, and (iii) *User privacy* by limiting what personal information is transmitted or retained.

Therefore, we propose a novel **M**ultitask **R**eflective **P**ivacy-preserving **L**arge **L**anguage **M**odel for Next POI Recommendation (MRP-LLM) to address the identified issues. As illustrated in Figure 1, MRP-LLM captures fine-grained preferences of both the user

[1]In our study, the terms 'POI' and 'location' are exchangeable.

and her neighbors through multitask prompting and self-reflection mechanisms, enhancing recommendation accuracy. Meanwhile, sensitive check-in data is protected during transmission to the RS.

Specifically, to effectively extract web users' behavioral patterns, a *Multitask Reflective Preference Extraction Module* first distills each user's categorical, temporal, and spatial preferences on POIs into a fine-grained preference knowledge base (KB). The *Neighbor Preference Retrieval Module* then identifies users' neighbors based on their historical distribution and social relationship, and retrieves and summarizes their preferences from the KB to inject collaborative signals. Subsequently, the *Multitask Next POI Recommendation Module* leverages the user's preferences with those of her neighbors to facilitate the next POI recommendation based on her check-in history. Moreover, to mitigate sensitive information leakage, raw check-in histories and social relationships are retained locally. A *Privacy Transmission Module* is designed to safeguard user data during transmission. Specifically, we propose a novel $\langle \rho, h \rangle$*-privacy* mechanism to protect users' check-in sequences. This mechanism is employed alongside other differential privacy techniques to secure sensitive user check-in history and social relationships. Its impact on user privacy is theoretically quantified and analyzed through the formal framework of expected privacy gain. As such, MRP-LLM exploits the LLMs to enhance recommendation accuracy while providing robust protection for user privacy.

In summary, the contributions of this study lie threefold. **(I)** We propose a novel multitask reflective LLM for next POI recommendation (MRP-LLM), which could enhance recommendation accuracy by reflectively extracting fine-grained user preferences and integrating collaborative signals from neighbors into the LLM recommendation process. **(II)** We propose a novel geolocation-based perturbation mechanism, $\langle \rho, h \rangle$*-privacy*, to protect users' check-in sequences. We theoretically analyze its privacy implications by quantifying the protection it provides using the formal framework of privacy gain. Building on this mechanism, we develop a comprehensive privacy-preserving scheme that safeguards all sensitive user data, including preferences, social relationships, and check-in history. **(III)** We conduct extensive experiments on three real-world datasets to verify the superiority of MRP-LLM over state-of-the-art methods. Specifically, when privacy protection is relaxed, it gains an average lift of 11.3% and 10.5% in ACC and MRR; with full-scale protection on users' sensitive data, it still achieves performance comparable to the best performing LLM-based next POI RS, limiting the performance drop to 1.4% and 1.6% w.r.t. ACC and MRR.

## 2 Related Work

**Next POI Recommendation**. Early studies utilized Markov chain to learn users' consecutive check-in behaviors [6, 42]. Later methods, such as ST-RNN [29] and STGN [71], incorporate spatial and temporal signals into recurrent neural networks [45]. The emergence of self-attention [55] inspires methods such as Deepmove [11], ATST-LSTM [19], STAN [35], and CFPRec [68] to achieve outstanding performance. Then, works like HMT-GRN [27], STHGCN [65] and DCHL [24] integrate graph models to enhance accuracy. Nevertheless, the conventional approaches face data sparsity.

**LLM-based Recommendation**. LLMs have received increasing attention in general recommendation due to their strong reasoning and knowledge transfer capabilities. Some works directly leverage ICL including zero-shot prompting [28] or chain-of-thought reasoning (CoT) [51, 57]. Others, such as KAR [64], LLMRec [63], and GaCLLM [9], integrate LLMs into traditional RSs to enrich user and item representations. Additionally, several works fine-tune LLMs specifically for recommendation tasks [2, 16, 30, 40]. LLMs have also shown strong potential in modeling sequential user-item interactions for next-item recommendation. In the ICL setting, both zero-shot [15] and few-shot prompting [18, 28] have been explored. Furthermore, ICL has been further enhanced with techniques such as self-reflection [54, 61] and retrieval-augmented generation [62]. Specifically in the context of next POI recommendation, works such as LLMMob [60], LLMMove [13], and AgentMove [12] leverage ICL based on users' check-in, temporal, and geographical data, but often lack fine-grained preference modeling and integration of collaborative signals. In contrast, fine-tuned approaches such as LLM4POI [25], SeCor [58], and GNPR-SID [56] are effective but resource-intensive. Moreover, none of the aforementioned methods protect the sensitive check-in histories and social relationships.

**Privacy-Preservation for Recommendation**. User privacy has been a key concern in conventional RSs with efforts focusing on cryptographic techniques [50, 72] and perturbation methods [10] to protect interaction history and side information. For conventional POI RSs, privacy protection typically targets users' check-in data using isolation [32], encryption [39], or perturbation [8], with some also preserving user preferences and social links [31], or geolocation-based data [20, 33]. For LLM-based models, although several methods have been proposed to mitigate both active attacks [38, 70] and passive leakages [22], for both fine-tuning [3, 46, 49] and inference stages [21, 34, 34, 37], privacy protection in LLM-based RSs, particularly for next POI recommendation, remains largely underexplored. Some works focus on protecting privacy only during inference, leaving data uploaded to the central server unprotected [5, 47]. A few efforts, such as RAPT [26], address this by perturbing input prompts. However, it relies on prompt tuning, which is resource-intensive and unsuitable for closed-source LLMs.

## 3 Methodology

**Notations**. Let $\mathcal{U} = \{u_1, u_2, ..., u_{|\mathcal{U}|}\}$, $\mathcal{L} = \{l_1, l_2, ...l_{|\mathcal{L}|}\}$, $\mathcal{C} = \{c_1, c_2, ...c_{|\mathcal{C}|}\}$ and $\mathcal{R} = \{r_1, r_2, ...r_{|\mathcal{R}|}\}$ denote the sets of users, POIs, POI categories, and POI regions respectively. Users' social relations are represented as a matrix $\mathbf{S}$. A check-in record $(u, l, t, r, c, d)$ denotes user $u$ visited POI $l$ at time $t$, where $l$ located in region $r$ is characterized by category $c$ and is $d$ km away from the user's previous check-in. For each user, we chronologically order her check-in records into check-in sequences. The most recently visited POI sequence is the *current POI sequence*, $\mathcal{L}_u^{cur}$, whereas the POIs visited previously form the *history POI sequence*, $\mathcal{L}_u^{hist}$. Both sequences comprise a list of POIs and check-in time, i.e., $\mathcal{L}_u = \{(l_{t_1}, t_1), (l_{t_2}, t_2), ..., (l_{t_k}, t_k)\}$. The corresponding category, region, and distance sequences are $\mathcal{C}_u = \{(c_{t_1}, t_1), (c_{t_2}, t_2), ..., (c_{t_k}, t_k)\}$, $\mathcal{R}_u = \{(r_{t_1}, t_1), (r_{t_2}, t_2), ..., (r_{t_k}, t_k)\}$, and $\mathcal{D}_u = \{(d_{t_1}, t_1), (d_{t_2}, t_2), ..., (d_{t_k}, t_k)\}$ [2].

[2]The complete notation table can refer to `'table_Notations'` in the project Github.

**Problem Statement**. The task of MRP-LLM is to recommend the next POI $l_{t_{k+1}}$ given users' current sequences $\mathcal{L}_u^{cur}, \mathcal{C}_u^{cur}, \mathcal{R}_u^{cur}, \mathcal{D}_u^{cur}$, history sequences $\mathcal{L}_u^{hist}, \mathcal{C}_u^{hist}, \mathcal{R}_u^{hist}, \mathcal{D}_u^{hist}$, and social relations $\mathbf{S}$.

**Framework Overview**. Figure 2 presents an overview of MRP-LLM. Users' data are strictly kept on their local devices. Various types of user data are transmitted to different modules on the RS server with privacy protection in place. Specifically, (1) Before recommendation, the *Multitask Reflective Preference Extraction Module* distills each user's fine-grained preferences from their category, region, and distance sequences via multitask preference probing and self-reflection. The preferences are stored in a preference knowledge base (KB) to enrich recommendation prompts. (2) During recommendation, the *Neighbor Preference Retrieval Module* first retrieves and summarizes neighboring users' preferences to inject the collaborative signal for recommendation prompts. (3) The *Multitask Next POI Recommendation Module* then recommends the next POI by aggregating both the user's and her neighbors' fine-grained preferences. (4) The *Privacy Transmission Module* mitigates users' check-in history and social relationship leakage during data uploading via perturbation techniques. In particular, we propose a comprehensive privacy protocol that incorporates a novel geolocation perturbation mechanism, $\langle \rho, h \rangle$-privacy, enabling effective recommendations without revealing users' check-in sequences or social relationships. We further conduct a rigorous theoretical analysis to quantify the expected privacy gain of the $\langle \rho, h \rangle$-privacy[3].

## 3.1 Multitask Reflective Preference Extraction

**User Model Schema**. This module aims to guide LLMs in understanding fine-grained user preferences to achieve more accurate recommendations. We first represent each user $u$ with an explicit, interpretable user model. Based on research [52], given the time-ordered check-in sequence $\mathcal{S}_u$, we define the user model as a set of five fine-grained preference factors:

$$\mathrm{UM}(u) = \{\mathbf{P}_u^{CT}, \mathbf{P}_u^{CT}, \mathbf{P}_u^{RT}, \mathbf{P}_u^{RT}, \mathbf{P}_u^{DT}\}. \tag{1}$$

- **Categorical transition preference** $\mathbf{P}_u^{CT}$: frequent category pairs $\langle c_i \rightarrow c_j \rangle$ visited consecutively (e.g., restaurant→ theatre).
- **Categorical temporal preference** $\mathbf{P}_u^{CT}$: time-conditioned category affinity (e.g., evening↦{bars}).
- **Regional transition preference** $\mathbf{P}_u^{RT}$: frequent region transitions $\langle r_i \rightarrow r_j \rangle$ (e.g., *Downtown* → *JurongEastRegion*).
- **Regional temporal preference** $\mathbf{P}_u^{RT}$: time-conditioned region affinity, (e.g., evening↦{Downtown District}).
- **Distance temporal preference** $\mathbf{P}_u^{DT}$: time-conditioned travel distance preference, (e.g., morning↦around 3 km).

As direct prompting could not effectively identify such fine-grained user preferences [13], we customize the prompts to probe user preferences step by step and refine iteratively[4].

**Multitask Preference Probing** We first use CoT to explicitly guide LLMs in capturing five types of user preferences across three subtasks. Specifically, after providing the task instruction (Prompt 1 in Appendix B), we present the LLM with user's current check-in category sequence $\mathcal{C}_u^{cur}$, her current check-in region sequence $\mathcal{R}_u^{cur}$, and her current check-in distance sequence $\mathcal{D}_u^{cur}$ and ask it to extract the user's categorical transition and temporal preferences (Prompt 2), regional transition and temporal preferences (Prompt 3), and distance temporal preferences (Prompt 4).

**User model update via reflective revision.** A single-pass extraction can miss evolving behaviors and is limited by context length for long histories. We therefore update $\mathrm{UM}(u)$ with self-reflection that uses prediction-based feedback to revise preference factors: **(1) Segment sampling.** To capture the recent evolution of user preferences, we first sample the most recent $m$ segments, each of length $n$, from the current sequence as examples. The self-reflection mechanism [36] is adopted to learn the recent preference changes from these segments. For category preference reflection, we sample $m$ most recent segments $\mathcal{C}_u^{cur'}$ from $\mathcal{C}_u^{cur}$. For each $\mathcal{C}_u^{cur'}$, we send the segment, excluding the last record, to the LLM and ask it to predict the final category (Prompt 5). We then provide the LLM with the ground truth category and guide it to rectify its previous conclusions on the user's categorical transition and temporal preferences (Prompt 6). Similar reflections are conducted to update the user's region preferences (Prompts 7&8), and distance preference (Prompts 9&10). **(2) Predict–reveal–revise.** Second, to integrate more information from the lengthy history sequence, we also sample $m$ segments of length $n$ each from the history sequence as examples to conduct the self-reflection. The segments are selected based on their contextual relevance to the recommendation task. Specifically, since the next POI recommendation task is to predict $l_{t_{k+1}}$ with current sequence $\{l_{t_1}, ...l_{t_{k-1}}, l_{t_{|\mathcal{C}|}}\}$, any segment in the history sequence whose second last check-in is $l_{t_{|\mathcal{C}|}}$ may have the highest context relevance to the recommendation task, followed by segments whose second last check-in is $l_{t_{k-1}}$, and so on. We feed these $m$ segments to LLMs and perform preference probing and self-reflection for each type of user preference. **(3) Write-back to KB.** Ultimately, the five types of user preferences will be stored in the fine-grained preference KB for retrieval during recommendation.

Overall, this mechanism explicitly models *user preference dynamics* by repeatedly correcting the profile using recent outcomes, while anchoring it with relevant historical evidence.

## 3.2 Neighbor Preference Retrieval

Solely relying on the user's own data fails to leverage global patterns. Thus, the Neighbor Preference Retrieval Module aims to leverage neighbors' preferences to introduce collaborative signals and further improve recommendation accuracy. We identify three types of neighbors. (1) *Geographical neighbors* means users having similar preferences on geographical regions, symbolized by the Kullback-Leibler (KL) divergence [23] of the regional check-in distribution, $RP(u) = \{\mathbb{P}(r_1), ..., \mathbb{P}(r_{|\mathcal{R}|})\}$,

$$D_{\mathrm{R}}(u_i, u_j) = \mathrm{KL}(RP(u_i) \parallel RP(u_j)); \tag{2}$$

(2) *Semantic neighbors* are users with similar category preferences, quantified by the KL divergence of check-in category distribution, $CP(u) = \{\mathbb{P}(c_1), ..., \mathbb{P}(c_{|\mathcal{C}|})\}$,

$$D_{\mathrm{C}}(u_i, u_j) = \mathrm{KL}(CP(u_i) \parallel CP(u_j)); \tag{3}$$

and (3) *Social neighbors* are the social network neighbors, i.e., $D_{\mathrm{S}}(u_i, u_j) = 1$ if $u_i$ and $u_j$ are friends and vice versa.

[3]Due to space limitations, supplementary diagrams, tables, and codes are available in our GitHub repository at https://anonymous.4open.science/r/MPR_LLM-8847.
[4]Detailed prompts could be found in `'detailed_prompts'` in the project Github.

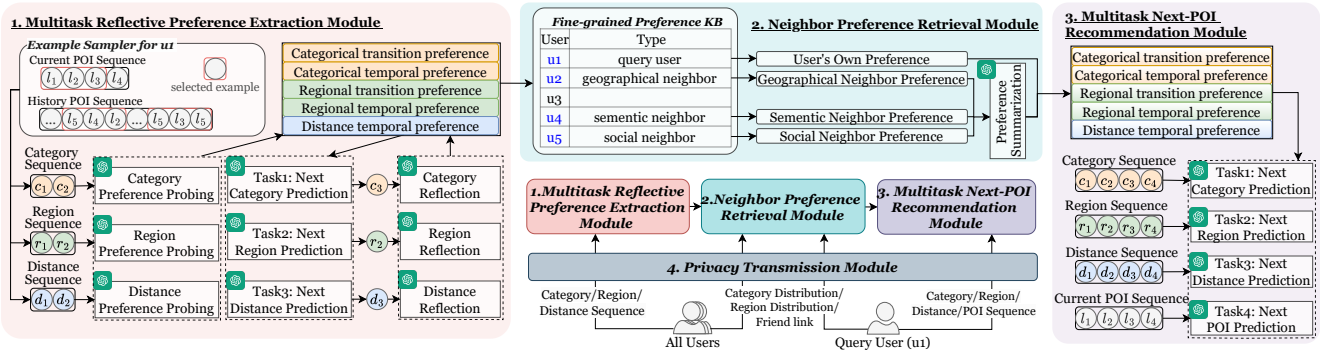


**Figure 2: The architecture of MRP-LLM. Taking $u_1$ as the recommendation query user.**

Based on $D_{\mathrm{R}}$, $D_{\mathrm{C}}$, and $D_{\mathrm{S}}$, it selects the closest neighbor of each type, retrieves their fine-grained preferences from the KB, and leverages LLMs to summarize each preference (Prompts 11-15).

## 3.3 Multitask Next POI Recommendation

This module aims to provide next POI recommendations by taking into account users' preferences on category, region, and distance. Specifically, we first predict the next preferable category, region, and distance in three separate tasks with the guidance of the user's and the neighbors' preferences (Prompts 16-18). The predictions on subtasks provide hints on different aspects of user preferences. Finally, the LLM is asked to consider the preference hints and select the most suitable POI candidates (Prompt 19). Additionally, the relative importance of each aspect and an explanation of the recommendation are required to enhance interpretability.

## 3.4 Privacy Transmission

This module seeks to preserve the privacy of four user data types uploaded to the three modules of MRP-LLM by employing differential privacy techniques [10], where details are described below.

(1) The Multitask Reflective Preference Extraction Module receives users' category, region, and distance sequences that could reveal users' preferences. Thus, we represent each record in a sequence by a one-hot vector and perturb it with optimized unary encoding (OUE) method [59]. OUE algorithm perturbs the $i$-th bit of the original vector $\mathbf{x}$ into $\mathbf{x}'$ with privacy budget $\epsilon$,

$$\mathbb{P}(\mathbf{x}'[i]=1)=\begin{cases}0.5, & \text{if } \mathbf{x}[i]=1\\ 1/(exp(\epsilon)+1), & \text{if } \mathbf{x}[i]=0.\end{cases} \tag{4}$$

(2) The Neighbor Preference Retrieval Module requires category and region check-in distribution to identify semantic and geographical neighbors, $RP(u)$ and $CP(u)$, and users' social relationship for neighbor retrieval. To protect the distribution, we inject Laplasian noise. Specifically, we set new distribution $P'(u)$ based on original distribution $P(u)$ and privacy budget $\epsilon$,

$$P'(u)=P(u)+Lap(\Delta f/\epsilon), \tag{5}$$

where the sensitivity $\Delta f$ in this case is 1. Furthermore, this module utilizes users' social relationship to seek their social neighbors. To protect user social links from exposure to unwanted personals or the recommender, we adopt the random flipping mechanism [14] to perturb each social link of the social relationship matrix $\mathbf{S}\in\mathbb{Z}^{|\mathcal{U}|\times|\mathcal{U}|}$. Specifically, we first draw a random probability $\phi\sim$ Uniform$(0,1)$ and then perturb the social link $s_{ij}$ between $u_i$ and $u_j$ to $s'_{ij}$ according to Equation 6, where $p/q\le exp(\epsilon)$,

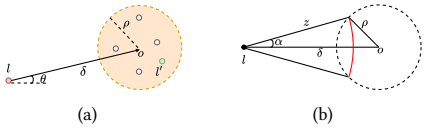


**Figure 3: Graphical Illustrations of $\langle\rho,h\rangle$-privacy Mechanism. (a) is an example of $\langle\rho,h\rangle$-privacy mechanism, where the original POI $l$ (red point) is first transformed into the orange circle, and then replaced by a random POI within the circle, $l'$ (green point); (b) is a geometric formulation of $\langle\rho,h\rangle$-privacy.**

$$s'_{ij}=\begin{cases}1, & \text{if } \phi\le p \text{ and } s_{ij}=1;\\ 0, & \text{if } \phi>p \text{ and } s_{ij}=1;\\ 0, & \text{if } \phi>q \text{ and } s_{ij}=0;\\ 1, & \text{if } \phi\le q \text{ and } s_{ij}=0.\end{cases} \tag{6}$$

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(3) Multitask Next POI Recommendation Module relies on users' privacy-sensitive POI sequences to generate accurate recommendations. To ensure both privacy protection and spatial utility, we propose a geolocation-specific privacy-preserving mechanism termed $\langle\rho,h\rangle$-*privacy*. Unlike the existing geolocation protection method [20], which perturbs only users' traveling distance, $\langle\rho,h\rangle$-privacy substitutes POIs with geographically proximate alternatives and simultaneously perturbs the POI ID $l$, the POI category $c$, and the traveling distance $d$. As illustrated in Figure 3 (a), for a user's check-in at location $l$, we fuzzify it by generating a virtual circle whose center is displaced by a distance $\delta$ at angle $\theta$ from $l$, and whose radius is $\rho$. The radius $\rho$ is dynamically adjusted based on the local check-in density $h$ within the circle to balance privacy and utility. Finally, the actual check-in location $l$ is replaced with a random alternative POI within the virtual circle to anonymize the user's true check-in. The detailed algorithm of $\langle\rho,h\rangle$-privacy is shown in Algorithm 1.

**Analysis of $\langle\rho,h\rangle$-privacy.** $\langle\rho,h\rangle$-privacy simultaneously perturbs three critical inputs of the multitask next POI recommendation module: $l$, $c$, and $d$. This motivates a quantitative evaluation of its privacy enhancement. To this end, we introduce an analytical framework grounded in the concept of expected privacy gain, derived from information entropy and inspired by the notion of privacy loss [7].

**Algorithm 1:** $\langle \rho, h \rangle$-Privacy for POI check-ins

**Input:** $l$ with coordinate $(x_l, y_l)$, privacy parameter $\epsilon$, $\delta_{\min}$,$\delta_{\max}$,
**Output:** replacement POI $l'$

```
δ ← random(δ_min, δ_max);
θ ← random(0, 2π);
o ← circle origin at (x_l + δ cos θ, y_l + δ sin θ);
h ← random(h_min, h_max);
ρ ← minimum radius containing h POIs;
l' ← a random POI in circle of origin o and radius ρ;
return l';
```

Definition 1. ***(Expected Privacy Gain)***. *Let $X$ be a variable, with $H(X_i)$ denoting the entropy of each record $X_i$ in the original dataset $\mathcal{D}$. Let $\mathcal{M}(X)$ represent a privacy-preserving mechanism that perturbs $X$, and define $H'(X_i) = H(\mathcal{M}(X_i))$ as the entropy of $X_i$ after perturbed data release. The Expected Privacy Gain of $\mathcal{M}$ is the expected increase in entropy across all records in $\mathcal{D}$: $PrivacyGain(\mathcal{M}(X)) = \mathbb{E}_{X_i \in \mathcal{D}}[H'(X_i) - H(X_i)]$.*

Intuitively, an entropy increase indicates greater uncertainty about a variable's true value, thereby providing a stronger protection of the underlying private information. Using this formulation, we assess the degree of privacy protection offered by $\langle \rho, h \rangle$-privacy.

Theorem 3.1. *Adapting POI IDs with $\langle \rho, h \rangle$-privacy results in an expected privacy gain of $\frac{1}{h_{\max} - h_{\min} + 1} \sum_{h=h_{\min}}^{h_{\max}} \ln h$.*

Proof. $\langle \rho, h \rangle$-privacy perturbs each POI $l_i$ by replacing it with another POI $l_i'$ sampled uniformly at random from a virtual circle containing $h$ POIs. When $h$ is fixed, the privacy gain for a single record is measured by the increase in Shannon entropy, $H(l_i'|h) - H(l_i) = -\sum^{h} \frac{1}{h} \ln\left(\frac{1}{h}\right) - 0 = \ln h$, as the original POI $l$ is deterministic (i.e., $H(l) = 0$). As $h$ is itself randomly drawn from $[h_{\min}, h_{\max}]$ under a discrete uniform distribution, the expected privacy gain across all records is:

$$\begin{aligned} \mathbb{E}_{l_i \in \mathcal{D}}[H'(l_i) - H(l_i)] &= \mathbb{E}_h \left[ \frac{1}{|\mathcal{D}|} \sum_{l_i \in \mathcal{D}} \left( H(l_i' \mid h) - H(l_i) \right) \right] \\ &= \frac{1}{h_{\max} - h_{\min} + 1} \sum_{h=h_{\min}}^{h_{\max}} \ln h. \end{aligned} \tag{7}$$

□

Theorem 3.2. *Adapting POI categories with $\langle \rho, h \rangle$-privacy yields an expected privacy gain of $\frac{1}{Z_c} \sum_{j=1}^{|C|} \frac{\ln j}{j} + \ln Z_c$, where $Z_c$ is the is the harmonic number of category rank.*

Proof. $\langle \rho, h \rangle$-privacy perturbs each record's category $c_i$ with replacement POI's category $c_i'$. We assume POI category follows Zipf's law due to its long-tail nature [48], i.e., for each category ranked $j$, the probability of observing $c_j$ is $P(c_j) = \frac{1/j}{Z_c}$, where $Z_c = \sum_{k=1}^{|C|} \frac{1}{k}$ serves as the normalization constant. For simplicity, we assume that the $h$ POIs in the virtual circle are sampled uniformly from the entire population and preserve the original categorical distribution. Thus, the expected privacy gain for all records is:

$$\begin{aligned} \mathbb{E}_{c_i \in \mathcal{D}}[H'(c_i) - H(c_i)] &= \frac{1}{|\mathcal{D}|} \sum_{c_i \in \mathcal{D}} (H(c_i') - H(c_i)) \\ &= -\sum_{j=1}^{|C|} \frac{1/j}{Z_c} \ln\left(\frac{1/j}{Z_c}\right) - 0 \\ &= \sum_{j=1}^{|C|} \frac{1/j}{Z_c} \ln j + \sum_{j=1}^{|C|} \frac{1/j}{Z_c} \ln Z_c \\ &= \frac{1}{Z_c} \sum_{j=1}^{|C|} \frac{\ln j}{j} + \ln Z_c. \end{aligned} \tag{8}$$

□

Theorem 3.3. *Adapting the travelling distance $d$ with $\langle \rho, h \rangle$-privacy results in a positive expected privacy gain lower bounded by $-\frac{2\alpha}{\pi\rho^2}\left[\left(\frac{b^2}{2}\ln b - \frac{b^2}{4}\right) - \left(\frac{a^2}{2}\ln a - \frac{a^2}{4}\right)\right]$, where $\alpha = \pi\rho/4\delta$, $a = \delta - \rho$, and $b = \delta + \rho$.*

Proof. In a consecutive check-in, $\langle \rho, h \rangle$-privacy perturbs both the departure point and the destination by introducing displacements $z_{\text{dep}}$ and $z_{\text{des}}$, respectively. For simplicity, we approximate the perturbed travel distance as the sum of the original distance and the two displacements, i.e., $d_i' = d_i + z_i^{\text{dep}} + z_i^{\text{des}}$. We first analyze the information increment induced by one side of the displacement, as illustrated in Figure 3(b). We assume the position of the replacement POI $l'$ is uniformly distributed in the virtual circle, i.e., the probability of $l'$ appearing at any point in the circle is $1/\pi\rho^2$. Thus, the probability density function of a displacement $z$ is the probability where $l'$ appears on the arc with angle $2\alpha$ and radius $z$, shown in the red line, i.e., $f(z) = 2\alpha z/\pi\rho^2$, where $z \in [\delta - \rho, \delta + \rho]$. To ensure $\int f(z) = 1$, $\alpha = \pi\rho/4\delta$. Let $a = \delta - \rho$ and $b = \delta + \rho$, the differential entropy of the displacement is therefore:

$$\begin{aligned} H(z) &= -\int_a^b \frac{2\alpha z}{\pi\rho^2} \ln\left(\frac{2\alpha z}{\pi\rho^2}\right) dz \\ &= -\int_a^b \frac{2\alpha z}{\pi\rho^2} \left(\ln\left(\frac{2\alpha}{\pi\rho^2}\right) + \ln z\right) dz \\ &= -\ln\left(\frac{2\alpha}{\pi\rho^2}\right) \int_a^b \frac{2\alpha z}{\pi\rho^2} dz - \int_a^b \frac{2\alpha z}{\pi\rho^2} \ln z \, dz \\ &= -\ln\left(\frac{2\alpha}{\pi\rho^2}\right) \cdot \frac{\alpha}{\pi\rho^2}(b^2 - a^2) \\ &\quad - \frac{2\alpha}{\pi\rho^2}\left[\left(\frac{b^2}{2}\ln b - \frac{b^2}{4}\right) - \left(\frac{a^2}{2}\ln a - \frac{a^2}{4}\right)\right], \end{aligned} \tag{9}$$

which is positive because $b > a$.

Since both the departure point and the destination undergo the same perturbation, i.e., $H(z^{\text{dep}}) = H(z^{\text{des}}) = H(z)$, the expected privacy gain for all records is:

$$\begin{aligned} \mathbb{E}_{d_i \in \mathcal{D}}[H'(d_i) - H(d_i)] &= \mathbb{E}_{d_i \in \mathcal{D}}[H(d_i + z_i^{\text{dep}} + z_i^{\text{des}}) - 0] \\ &= \frac{1}{|\mathcal{D}|} \sum_{d_i \in \mathcal{D}} H(z_i^{\text{dep}} + z_i^{\text{des}}) > H(z), \end{aligned} \tag{10}$$

where the last inequality is derived because differential entropy increases over the convolution of two i.i.d. variables, i.e., $H(z_i^{\text{dep}} + z_i^{\text{des}}) > \max(H(z_i^{\text{dep}}), H(z_i^{\text{des}}))$. □

Theorem 3.4. *$\langle \rho, h \rangle$-privacy introduces an expected privacy gain upper bounded by $H(1/h) + H(2z/4\delta\rho)$ to MRP_LLM.*

Proof. We treat the recommendation engine of MRP-LLM as a black-box function, denoted by LLM(·), which takes four random variables as input: POI ID $l$, POI category $c$, POI region $r$, and traveling distance $d$. Under $\langle \rho, h \rangle$-privacy, $l$, $c$, and $d$ are perturbed while $r$ is largely preserved as the replacement POIs are selected from a nearby geographical location. Based on the chain rule of entropy, the upper bound of $\langle \rho, h \rangle$-privacy can be derived by:

$$\begin{aligned}
&\mathbb{E}_{i\in\mathcal{D}}[H(\mathrm{LLM}(l'_i, c'_i, r_i, d'_i)) - H(\mathrm{LLM}(l_i, c_i, r_i, d_i))] \\
&< \mathbb{E}_{i\in\mathcal{D}}[H(l'_i, c'_i, r_i, d'_i)] \\
&= \mathbb{E}_{i\in\mathcal{D}}[H(l'_i) + H(c'_i \mid l'_i) + H(r_i \mid l'_i, c'_i) + H(d'_i \mid l'_i, c'_i, r_i)] \\
&< \frac{1}{|\mathcal{D}|} \sum_{d_i\in\mathcal{D}} (H(l'_i) + 0 + 0 + H(d'_i)) \\
&= H(1/h) + H(2z/4\delta\rho),
\end{aligned} \tag{11}$$

where the last inequality is due to the POI category $c'$ is dependent on $l'$ and the unperturbed region $r$ is deterministic, i.e., $H(c' \mid l') = 0$ and $H(r \mid l', c') = 0$. □

In summary, the entire process for MRP-LLM to perform next POI recommendation task is summarized in Algorithm 2. First, users upload their perturbed category, region, and distance sequences, together with their perturbed category, region distributions, and social relationship (lines 3-5). MRP-LLM then samples $m$ segments from uploaded sequences based on time and contextual relevance (line 15) and uses the segments to perform multiple rounds of multitask preference probing and dynamic preference self-reflection (lines 16-18). The resulting users' preferences are stored in the fine-grained preference KB (line 7). Second, when a user initiates a recommendation quest, MRP-LLM identifies her neighbors based on geographical, semantic, and social closeness, retrieves their preferences, and utilizes LLM to summarize their fine-grained preferences (lines 9-10). Finally, the user's and her neighbors' preferences serve as important hints for LLM to perform the multitask next POI recommendation based on the user's perturbed current category, region, distance, and POI sequence (lines 11-13).

**Algorithm 2:** Recommendation Process of MRP-LLM

**Input:** $\mathcal{U}$, $\mathcal{L}_u$, $\mathcal{C}_u$, $\mathcal{R}_u$, $\mathcal{D}_u$
**Output:** Next POI recommendation list $\mathbf{l}$
`// Before recommendation request`
1 Input general instruction (*Prompt 1*);
2 **for** $u$ *in* $\mathcal{U}$ **do**
3 Collect $\mathcal{C}'_u$, $\mathcal{R}'_u$, $\mathcal{D}'_u$ perturbed as Equation 4;
4 Collect $RP'(u)$, $CP'(u)$ perturbed as Equation 5;
5 Collect $\mathbf{S}'$ perturbed as Equation 6;
6 $Pref(u) \leftarrow$ `PreferenceExt(`$\mathcal{C}'_u$, $\mathcal{R}'_u$, $\mathcal{D}'_u$`)`;
7 Store $Pref(u)$ into fine-grained KB;
`// Upon each recommendation request`
8 **for** $u$ *in* $\mathcal{U}$ **do**
9 Find $u$'s geographical, semantic, and social neighbors;
10 Summarize neighbor preferences (*Prompt 5*);
11 Collect POI sequences perturbed by Algorithm 1;
12 $\mathbf{l} \leftarrow$ next POI recommendation (*Prompts 6-7*);
13 **return** $\mathbf{l}$;
14 **Function** `PreferenceExt(`$\mathcal{C}_u$, $\mathcal{R}_u$, $\mathcal{D}_u$`)`:
15 Select $m$ segments from $\mathcal{C}_u$, $\mathcal{R}_u$, $\mathcal{D}_u$;
16 **for** *each example* **do**
17 $Pref(u) \leftarrow$preference extraction (*Prompt 2*);
18 $Pref(u) \leftarrow$preference reflection (*Prompts 3-4*);
19 **return** $Pref(u)$;

## 4 Experiments and Analysis

We conducted experiments on three real-world datasets to answer the following four research questions[5]. (**RQ1**) How does the recommendation accuracy of MRP-LLM compare to existing conventional and LLM-based next POI RSs? (**RQ2**) How does each component affect the performance of MRP-LLM? (**RQ3**) How do hyperparameters affect the performance of MRP-LLM? (**RQ4**) How reasonable and interpretable is the output of MRP-LLM? (**RQ5**) How robust is MRP-LLM to prompt and hallucination?

### 4.1 Experimental Setup

**Datasets.** We utilized the Foursquare dataset [66] in three cities, namely Singapore (SIN), New York (NY), and Phoenix (PHO), constituting user check-in history, POI geographical location, and user friendships[6]. Following existing works [69], we preprocess the dataset by applying 5-core filtering. For each user, we split the check-ins by day and filter out users with fewer than 3 sequences. The sequences are then chronologically split into training, validation, and test sets with a ratio of 8:1:1. Since MRP-LLM performs ICL without the need for model training, the training and validation sets are used as history sequences. The test set serves as the current sequence with the last check-in as ground truth.

**Evaluation Metrics.** In line with SOTA approaches [13, 60], we use Accuracy (ACC@K) and Mean Reciprocal Rank (MRR) to evaluate all methods. In general, higher metric values indicate better ranking performance. To ensure robust comparisons, we repeated each evaluation 10 times and reported the average result.

**Compared Baselines.** We compare MRP-LLM with baselines for conventional next POI RSs (1-7) and LLM-based next POI RSs (8-10), (1) ***MostPop*** recommends the most popular POIs; (2) ***Dist*** selects the nearest POIs; (3) ***BPRMF*** [43] performs matrix factorization on check-ins; (4) ***STRNN*** [29] incorporats check-in time and distance into RNN; (5) ***STAN*** [35] utilizes bi-attention architectures to exploit the check-in spatiotemporal correlation; (6) ***STHGCN*** [65] leverages POI, category, and spatiotemporal differences to perform hypergraph learning. (7)***HMST*** [41] leverages hyperbolic rotations to model hierarchical mobility patterns and transition semantics; (8) ***LLMMob*** [60] performs zero-shot prompting based on users' history and current sequences; (9) ***LLMMove*** [13] performs zero-shot prompting based on users' check-in sequences, distance, and transition patterns. (10) ***AgentMove*** [12] employs chain-of-thought zero-shot prompting by incorporating users' mobility patterns, urban structure knowledge, and the check-in histories of friends.

**Implementation Details.** For LLM-based methods, we use *gpt-3.5-turbo* as the engine for recommendation. For conventional RSs, experiments are conducted on a machine equipped with an Intel i7-10700 CPU @ 2.90GHz, 32 GB of RAM, and an NVIDIA RTX 2080 GPU. For each evaluation, we randomly sample 100 POIs containing the ground truth as candidate sets. For all conventional next POI RSs, we either empirically find out their best parameter settings or follow the suggested settings in the original papers[7].

[5]Our codes are available in `'source_code'` in the project Github.
[6]Detailed statistics can be found in `'data_statistics'` in the project Github.
[7]Detailed parameter settings can refer to `'hyperparameters'` in the project Github.

**Table 1: Performance comparison of conventional (rows 1-7) and LLM-based baselines (rows 8-10) with our methods without privacy preservation/with $\langle \rho, h \rangle$-privacy or DP (rows 11-13) on SIN, NY, and PHO. We repeat each evaluation 10 times and report the mean ± std. The improvement (row 14) is verified with a paired t-test, whose $p < 0.05$ is marked with * and $p < 0.005$ with **.**

| | | SIN | | | | NY | | | | PHO | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ACC@1 | ACC@5 | ACC@10 | MRR | ACC@1 | ACC@5 | ACC@10 | MRR | ACC@1 | ACC@5 | ACC@10 | MRR |
| Conventional | MostPop | 0.0550±0.003 | 0.2050±0.005 | 0.2600±0.006 | 0.0973±0.004 | 0.0300±0.002 | 0.1450±0.004 | 0.1950±0.005 | 0.0729±0.003 | 0.1400±0.004 | 0.2550±0.006 | 0.3000±0.007 | 0.1960±0.005 |
| | Dist | 0.1700±0.005 | 0.2700±0.006 | 0.3150±0.007 | 0.2452±0.005 | 0.0700±0.004 | 0.1650±0.005 | 0.2300±0.006 | 0.1052±0.004 | 0.2050±0.006 | 0.3050±0.007 | 0.3600±0.008 | 0.2766±0.006 |
| | BPRMF | 0.2100±0.006 | 0.3050±0.007 | 0.3850±0.008 | 0.2477±0.006 | 0.1600±0.005 | 0.2550±0.006 | 0.3550±0.007 | 0.2077±0.005 | 0.2400±0.007 | 0.3500±0.008 | 0.4250±0.009 | 0.3216±0.007 |
| | STRNN | 0.2050±0.007 | 0.3500±0.008 | 0.4100±0.009 | 0.2764±0.007 | 0.1750±0.006 | 0.3000±0.007 | 0.3900±0.008 | 0.2317±0.006 | 0.2850±0.008 | 0.4000±0.009 | 0.4850±0.010 | 0.3742±0.008 |
| | STAN | 0.6000±0.009 | 0.6800±0.008 | 0.7400±0.010 | 0.6373±0.009 | 0.4750±0.008 | 0.5800±0.008 | 0.6550±0.009 | 0.5135±0.008 | 0.6200±0.009 | 0.6950±0.008 | 0.7750±0.010 | 0.6587±0.009 |
| | STHGCN | 0.6150±0.010 | 0.6550±0.009 | 0.7600±0.010 | 0.6313±0.009 | 0.5000±0.009 | 0.5850±0.008 | 0.6500±0.009 | 0.5346±0.008 | 0.6300±0.010 | 0.6800±0.009 | 0.7950±0.010 | 0.6662±0.009 |
| | HMST | 0.6200±0.010 | 0.6600±0.009 | 0.7650±0.011 | 0.6402±0.010 | 0.5000±0.009 | 0.5750±0.009 | 0.6500±0.010 | 0.5273±0.009 | 0.6400±0.011 | 0.6800±0.010 | 0.7900±0.011 | 0.6437±0.010 |
| LLM-based | LLMMob | 0.3700±0.013 | 0.4950±0.012 | 0.5400±0.013 | 0.4231±0.011 | 0.3100±0.012 | 0.4550±0.011 | 0.4800±0.012 | 0.3379±0.010 | 0.3950±0.014 | 0.5650±0.013 | 0.6700±0.014 | 0.4436±0.012 |
| | LLMMove | 0.3950±0.014 | 0.4700±0.013 | 0.5500±0.014 | 0.4428±0.012 | 0.3000±0.013 | 0.4650±0.012 | 0.5100±0.013 | 0.3611±0.011 | 0.4100±0.015 | 0.5750±0.014 | 0.6800±0.015 | 0.4652±0.013 |
| | AgentMove | 0.5350±0.012 | 0.6000±0.011 | 0.6300±0.012 | 0.5559±0.010 | 0.4200±0.011 | 0.5150±0.010 | 0.5600±0.011 | 0.4735±0.009 | 0.5400±0.013 | 0.6300±0.012 | 0.7100±0.013 | 0.5903±0.011 |
| | **MR-LLM** (w/o privacy) | **0.6150**±0.011 | **0.6550**±0.010 | **0.7050**±0.011 | **0.6276**±0.009 | **0.4750**±0.010 | **0.5750**±0.009 | **0.6400**±0.010 | **0.5124**±0.008 | **0.6350**±0.011 | **0.6700**±0.010 | **0.7500**±0.011 | **0.6513**±0.009 |
| | **MRP-LLM** ($\langle \rho, h \rangle$-privacy) | 0.5000±0.012 | 0.5750±0.010 | 0.6300±0.011 | 0.5388±0.009 | 0.4150±0.011 | 0.5200±0.010 | 0.5700±0.011 | 0.4679±0.009 | 0.5350±0.012 | 0.6250±0.011 | 0.7000±0.012 | 0.5879±0.010 |
| | MR-LLM (with DP) | 0.3700±0.014 | 0.4050±0.013 | 0.4500±0.014 | 0.3926±0.012 | 0.3450±0.013 | 0.3750±0.012 | 0.4200±0.013 | 0.3625±0.011 | 0.4300±0.015 | 0.5550±0.014 | 0.6100±0.015 | 0.4817±0.013 |
| | **Imp.** for MRP-LLM | -6.5% | -4.2% | 0.0% | -3.1% | -1.2% | 1.0% | 1.8% | -1.2% | -1.0% | -1.0% | -1.4% | 0.0% |

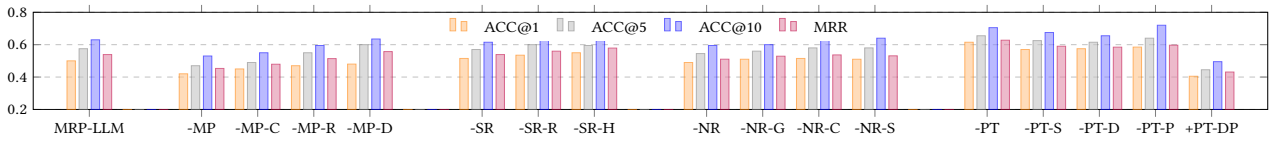


**Figure 4: Ablation study results on SIN.**

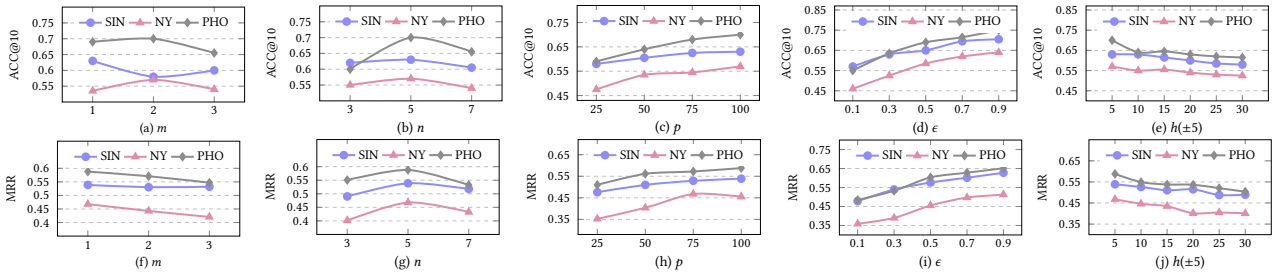


**Figure 5: Parameter sensitivity analysis of key hyper-parameters on ACC@10 and MRR**

## 4.2 Experiment Results and Analysis

**Model Performance Comparison (RQ1).** Table 1 shows the performance of all baselines and our methods across the three datasets. Underlined scores indicate the best results among conventional baselines, while double-underlined scores denote the best among LLM-based baselines. Our methods, MR-LLM (without privacy) and MRP-LLM (with privacy), are highlighted in bold. To fairly compare the effectiveness of $\langle \rho, h \rangle$-privacy, we apply standard differential privacy on user's check-in POIs [4] shown in the row MR-LLM (w d.p.). The improvement of MR-LLM against the best-performing LLM-based next POI RS is shown in the last row. Several observations follow. **(1)** SOTA conventional POI recommendation methods, such as STAN, STHGCN, and HMST trained with ***full-size data***, achieve the highest accuracy. We explicitly acknowledge this and do not claim to surpass classical models on pure ranking metrics. Instead, our goal is to explore the potential of LLMs for next-POI recommendation under user-centric requirements: (i) avoiding resource-intensive training/fine-tuning on full-scale user data by leveraging strong inference-time capabilities of LLMs, (ii) enabling more interpretable recommendations through explicit preference factors and preference-grounded explanations, and (iii) providing a framework likely to benefit from continued improvements in foundation models, which may narrow the current gap over time. Our method (MR-LLM), having achieved the above benefits, still achieved comparable performance with computational-demanding conventional methods, limiting the average decrease to 3.6% and 2.6% in terms of ACC and MRR. With the privacy-preserving transmission enabled, our method (MRP-LLM) achieves strong accuracy compared to other LLM-based methods while significantly reducing the exposure of sensitive user information. Our approach is most suitable when (i) rapid iteration is needed without retraining, (ii) cold-start/sparse behavioral data benefits from LLMs' cross-domain knowledge, (iii) explanations/controllability are required, or (iv) personalization must operate under privacy constraints. **(2)** For LLM-based methods, AgentMove performs better than the others by leveraging external knowledge on urban structure and friends' interactions. Our MR-LLM greatly outperforms AgentMove with an average lift of 11.3% and 10.5% w.r.t. ACC and MRR (see line 13). Even with privacy preservation, MRP-LLM could achieve similar performance compared to LLMMove, with a slight drop of 1.4% and 1.6% w.r.t. ACC and MRR (see line 14). **(3)** MR-LLM outperforms other LLM-based next POI recommenders and achieves performance comparable to conventional models trained on full-size data. This underscores the effectiveness of leveraging tailored ICL strategies for precise spatial-temporal preference extraction and integrating KB to infuse collaborative signals. Meanwhile, MRP-LLM

achieves the best balance between privacy and utility, achieving on-par performance with non-privacy-preserving LLM-based SOTAs. **(4)** MR-LLM with $\langle \rho, h \rangle$-privacy significantly outperforms standard DP across all datasets, particularly on those with larger POI volumes. We attribute this improvement to the ability of $\langle \rho, h \rangle$-privacy to preserve geographical proximity when perturbing locations, which is critical for maintaining spatial signals in POI recommendation.

**Component Effectiveness Analysis (RQ2).** We examine the efficacy of different components of MRP-LLM by removing each of them. **First**, we remove multitask preference probing (-MP). Within this component, we also remove each subtask to examine their impacts, i.e., category (-MP-C), region (-MP-R), and distance (-MP-D) probing, respectively. **Second**, we remove the self-reflective component (-SR). Within the component, we also remove each type of segment used for reflection, i.e., segments drawn from recent sequences (-SR-R) and history sequences (-SR-H). **Third**, We remove the Neighbor Preference Retrieval Module (-NR). Within this component, we further remove each type of neighbor, i.e., geographical (-NR-G), semantic (-NR-C), and social neighbors (-NR-S). **Last**, We remove the Privacy Transmission Module (-PT). Within this module, we remove perturbations on each type of data: the category, region, and distance sequences (-PT-S), the category and region distributions (-PT-D), and the POI sequences (-PT-P). Moreover, we replace $\langle \rho, h \rangle$-privacy with standard exponential noise differential privacy (+PT-DP) to validate our mechanism's efficacy.

Figure 4 presents the experimental results of the component effectiveness analysis on the SIN dataset, with similar trends observed on NY and PHO[8]. It reveals several key observations. **(1)** MP improves model accuracy. Among them, the subtask on category has the largest impact on most occasions, followed by region and distance. **(2)** SR enhances the performance, where temporal and contextual segments can lead to comparable improvements. **(3)** NR can effectively enhance recommendation accuracy, where the geographical neighbors have the highest contribution, followed by semantic and social neighbors; **(4)** In PT, the protection on POI sequences degrades accuracy the most, while perturbing distributions have the least impact on performance. **(5)** $\langle \rho, h \rangle$-privacy outperforms standard differential privacy with exponential mechanisms as it can preserve the geolocation proximity.

**Parameter Sensitivity Analysis (RQ3).** Figures 5 lists the impact of essential hyper-parameters, including the segment number $m$ and length $n$, the preference extraction participation rate $p$, i.e., the percentage of users participating in multitask preference extraction and saving preferences to the KB for neighbor retrieval, the privacy budget $\epsilon$, and the POI density $h$ in the fuzzified virual circle on three datasets w.r.t. ACC@10 and MRR[9]. Several findings are noted. **(1)** There is no obvious consistent trend on the impact of $m$ on accuracy across all datasets. Considering efficiency, we believe choosing one for each type of segment is optimal. **(2)** The optimal length of each segment is $n = 5$, which can provide sufficient context and avoid irrelevant noise. **(3)** MRP-LLM performs the best when $p = 100\%$. However, it could achieve comparable accuracy even when only 50% of users upload their preferences, indicating its robustness. **(4)** A higher $\epsilon$ results in less noise added to user data and increased accuracy, illustrating the privacy-utility trade-off. **(5)** A higher $h$ leads to greater privacy gain. However, when $h$ exceeds a certain threshold, it causes a sharp drop in recommendation performance.

**Recommendation Case Study (RQ4).** Our offline experiments support claims about recommendation quality, robustness, and the impact of individual components and privacy-preserving settings. However, due to resource constraints, we do not conduct a user study or online A/B test for interpretability-related claims. We therefore show a case study as qualitative evidence of how MRP-LLM could generate meaningful explanations for its recommendation. We performed the case study at each stage of a real recommendation process by randomly sampling a user on the PHO dataset. **(1)** During multitask preference probing (Stage 1), the user's preferences can be accurately captured. For example, the model could effectively capture the user's category transition preferences (e.g., 'pet services→movie theatres') and category temporal preferences (e.g., Sunday morning going to pet services). **(2)** During dynamic preference self-reflection (Stage 2), the user's preferences are updated by reflecting on examples. For instance, when the model shows that the user visited a restaurant after a pet service, the 'pet services→restaurants' transition is added to her preference profile (shown in red). **(3)** During neighbor preference summarization (Stage 3), neighbors' preferences are correctly aggregated. **(4)** During next POI recommendation (Stage 4), the model correctly infers the next category 'Restaurants' by considering both the user's category transition preferences and the collaborative signals from neighbors. The model then weighs different user preferences to provide accurate and well-reasoned recommendations on the ground truth POI (shown in red).

**Stage 1: Multitask Preference Probing**

**Prompt:** ⟨Prmopt 1⟩. Given the user's Category sequence: {(Pet Services, Sun, 10am), (Movie Theaters, Sun, 11am), (Coffee Shop, Sun, 2pm), (Grocery, Sun, 4pm), (Bars, Sun, 8pm)} what is the user's categorical transition preference? Considering: what are the 'category pairs' the user usually visits consecutively? (format:category-category,...)
**Output:** ***{Pet Services-Movie Theaters, Movie Theaters-Coffee Shop, Coffee Shop-Grocery, Grocery-Bars}***.
**Prompt:** What is the user's categorical temporal preference? Considering: what are the 'categories' the user visits at a certain time (day/ hour)? (format:{time: [categories]})
**Output:** ***{morning: [Pet Services, Movie Theaters], afternoon: [Coffee Shop, Grocery], evening: [Bars]}***.

**Stage 2: Dynamic Preference Self-Reflection**

**Prompt:** The user has visited categories {(Coffee Shop, Mon, 8am), (Pet Services, Mon, 10am). Now is Mon at 12pm, based on the user's categorical transition preference and categorical temporal preference, predict users' next most likely visiting 'Category'. (format: Category)
**Output:** ***Movie Theatres.***
**Prompt:** The user actually visited {Restaurants}. Based on this, what is the new insight you can get for the user's categorical transition preference? Generate the updated categorical transition preference. (format:{category-category,...})
**Output:** ***{Pet Services-Movie Theaters, Movie Theaters-Coffee Shop, Coffee Shop-Grocery, Grocery-Bars, Coffee Shop-Pet Services, Pet Services-Restaurants}***.

**Stage 3: Neighbor Preference Summarization**

**Prompt:** The users' geographical neighbors' categorical transition preferences are {Restuarants-Bars, Pet Services-Coffee Shop}; the users' semantic neighbors'

[8]Results can refer to 'ablation_study' in the project Github.
[9]ACC@1 and ACC@5 can refer to 'sensitivity_study' in the project Github.
[10]Detailed breakdown can refer to 'efficiency_analysis' in the project Github.

**Table 2: Robustness analysis of the proposed framework on three datasets. Consistency is measured by repeated prompting under identical inputs. Hallucination rate denotes the proportion of outputs failing validation.**

| Dataset | Schema-level Consistency (%) | Hallucination Rate (%) |
|---|---|---|
| SIN 1 | 93.4 | 4.8 |
| NY 2 | 91.7 | 5.3 |
| PHO 3 | 94.1 | 4.1 |

categorical transition preferences are {Coffee Shop-Pet Services, Pet Services-Restaurants}; the users' social neighbors' categorical transition preferences are {Department Store-Clothing, Restaurants-Bars, Grocery-Bakeries}. Summarize the neighbors' categorical transition preferences by considering their commonalities. (format: {category-category,...}).
**Output:** ***{Restaurants-Bars, Pet Services-Coffee Shop, Pet Services-Restaurants, Department Store-Clothing, Grocery-Bakeries}.***

**Stage 4: Next POI Recommendation**

**Prompt:** Now is {Mon} at {12pm}, based on the users' current category sequence {Coffee Shop, Mon, 8am), (Pet-Services, Mon, 10am)}, his own categorical transition preference and categorical temporal preference, and his neighbors' categorical transition preference and categorical temporal preference, predict the user's next most likely visiting 'category'. (format: category)
**Output:** ***Restaurants.***
⟨Next Region and Distance Prediction⟩
**Prompt:** Candidate POIs: (format:(Name, Category, Region, Distance)): {(The Melt, Restaurants, r3, 1km), (Crescent Ballroom, Bar, r3, 2km), (Pizzeria Bianco, Restaurants, r1, 12km), ...} Given users' current check-in sequence {(Espresso Bar, Coffee Shop, r3, Mon, 8am), (Petbar Phoenix, Pet Services, r3, Mon, 8am)}, recommend 10 POIs from the candidate set, considering his next likely visiting category {Restaurants}, region{r3}, and distance{2km}. State the reason for each recommendation and rank the importance of category, region, and distance preferences. (format: {POI: reason; [importance ranking]})
**Output:** ***The Melt: Matches next likely category 'Restaurants', in preferred region 'r3', and within 2 km distance; [Category > Region > Distance].***

**Robustness Analysis (RQ5).** Because our framework is prompting-based, we evaluate the robustness of our design via (i) consistency of preference extraction under repeated prompting and (ii) resistance against hallucination. For (i), we run a preference extraction prompt three times under identical input conditions for each user. We then compute *schema-level consistency rate*, which measures the proportion of the extracted preferences remaining the same across 3 runs for the same user. For (ii), we measure the hallucination rate for all the POI recommendations, defined as the proportion of generated POIs that cannot be confidently aligned with a valid candidate POI. The results are reported in Table 2. We observe that the proposed framework achieves high schema-level consistency across repeated prompting (>90%), and the hallucination rate remains low (around 5%). These results suggest that the proposed method is robust enough for practical use and that the validation step is effective in mitigating common LLM reliability issues.

**Limitations and Future Work.** Since MRP-LLM integrates several techniques to support ICL, including multitask preference extraction, dynamic self-reflection, and neighbor preference retrieval, it significantly improves accuracy at the expense of efficiency. Empirical results show that its inference time is approximately twice that of AgentMove, primarily due to the preparation of user profiles and the overhead introduced by privacy protocols. Additionally, the monetary cost of invoking LLM APIs is around 1.5 times higher, attributed to fine-grained preference probing and the reasoning explanation generation, posing challenges for future efficiency optimization. MRP-LLM achieves comparable overall runtime to deep models by avoiding iterative backpropagation, and its recommendation time is similar to other LLM-based RSs[10].

While our approach demonstrates that the above techniques improve next-POI recommendation and explanation quality, several extensions are promising. First, we will improve efficiency by reducing prompt length and inference cost via model distillation and retrieval caching. We will also explore tighter integration of users' short-term intent and long-term preferences with a more compact preference schema and user modeling. Second, we will explore adaptive privacy–utility trade-offs, including dynamic privacy budgets and stronger privacy auditing to quantify potential information leakage under different personalization settings. Third, we will conduct rigorous user-centric evaluations for interpretability and controlability, with a controlled study comparing our preference-grounded explanations to baselines, measuring helpfulness, clarity, decision confidence, and privacy preferences.

## 5 Conclusion

In this paper, we propose the novel MRP-LLM, which is tailored to perform a more accurate next POI recommendation and preserve check-in privacy with LLMs. Specifically, it first sets multiple subtasks and leverages a self-reflection mechanism to iteratively extract and refine users' fine-grained POI-related preferences. During recommendation, it retrieves neighbor users' signals to facilitate the next POI recommendation. Besides, all sensitive data are kept on local devices and transmitted with differential privacy. Extensive experiments on three real-world datasets demonstrate that MRP-LLM could effectively enhance LLMs' performance on the next POI recommendation task and achieve the best privacy-utility balance.

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